TS1 - Marlene Koss

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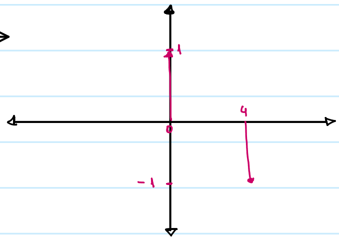
$\boxed{TS1}$

$h[n] = \delta[n] - \delta[n-4]$



$y[n] = x[n] * h[n] \Rightarrow Y(\omega) = X(\omega)H(\omega)$

$$H(\omega) = \sum_{n=-\infty}^{\infty} (\delta[n] - \delta[n-4]) e^{-j\omega n} = 1 - e^{-j4\omega}$$

a) $x_a[n] = \cos(\omega_0 n T_s) = \frac{1}{2}\left(e^{j\omega_0 n T_s} + e^{-j\omega_0 n T_s}\right)$

$$X_a(\omega) = \pi \sum_{\ell=-\infty}^{\infty} [\delta(\omega - \omega_0 T_s - 2\pi\ell) - \delta(\omega - \omega_0 T_s - 2\pi\ell)]$$

tomando solo el primer periodo de repetición

$$X_a(\omega) = \pi[\delta(\omega - \omega_0 T_s) - \delta(\omega - \omega_0 T_s)]$$

$$Y_a(\omega) = \pi \sum_{\ell=-\infty}^{\infty} [\delta(\omega - \omega_0 T_s - 2\pi\ell) - \delta(\omega - \omega_0 T_s - 2\pi\ell)] \cdot (1 - e^{-j4\omega})$$

corrimiento en el dominio del tiempo

$$= \pi \sum_{\ell=-\infty}^{\infty} [\delta(\omega - \omega_0 T_s - 2\pi\ell) - \delta(\omega - \omega_0 T_s - 2\pi\ell)] - \pi e^{-j4\omega} \sum_{\ell=-\infty}^{\infty} [\delta(\omega - \omega_0 T_s - 2\pi\ell) - \delta(\omega - \omega_0 T_s - 2\pi\ell)]$$

$$\Rightarrow y_a[n] = \cos(\omega_0 T_s n) - \cos(\omega_0 T_s (n-4))$$

$$= \cos(\omega_0 T_s n) - \cos(\omega_0 T_s n)\cos(\omega_0 T_s 4) - \operatorname{sen}(\omega_0 T_s n)\operatorname{sen}(\omega_0 T_s 4)$$

$$= \underbrace{(1 - \cos(\omega_0 T_s 4))}_{a} \cos(\omega_0 T_s n) - \underbrace{\operatorname{sen}(\omega_0 T_s 4)}_{b} \operatorname{sen}(\omega_0 T_s n)$$

quiero que: $\underbrace{A\cos(\omega_0 n T_s + \varphi)}_{} = a\cos(\omega_0 T_s n) - b\,\operatorname{sen}(\omega_0 T_s n)$

$= A\cos(\omega_0 n T_s)\cos(\varphi) - A\operatorname{sen}(\varphi)\operatorname{sen}(\omega_0 n T_s)$

$$\Rightarrow \begin{cases} A\cos(\varphi) = (1 - \cos(\omega_0 T_s 4)) & (1) \\ A\operatorname{sen}(\varphi) = +\operatorname{sen}(\omega_0 T_s 4) & (2) \end{cases}$$

elevando ambos al cuadrado:

$A^2\cos^2(\varphi) = 1 - 2\cos(\omega_0 T_s 4) + \cos^2(\omega_0 T_s 4)$

$A^2\operatorname{sen}^2(\varphi) = \operatorname{sen}^2(\omega_0 T_s 4)$

$\operatorname{sen}^2(\alpha) = \frac{1 - \cos(2\alpha)}{2}$

$(1)^2 + (2)^2:\ A^2 = 2 - 2\cos(\omega_0 T_s 4) = 4\operatorname{sen}^2(2\omega_0 T_s)$

$|A| = |2\operatorname{sen}(2\omega_0 T_s)|$

$\operatorname{sen}(2\alpha) = 2\operatorname{sen}(\alpha)\cos(\alpha)$ $\qquad \cot(\alpha) = \operatorname{tg}(\frac{\pi}{2} - \alpha)$

$$\frac{(2)}{(1)}:\ \operatorname{tg}(\varphi) = \frac{\operatorname{sen}(\omega_0 T_s 4)}{1 - \cos(\omega_0 T_s 4)} = \frac{\cancel{2}\,\cancel{\operatorname{sen}(2\omega_0 T_s)}\cos(2\omega_0 T_s)}{\cancel{2}\operatorname{sen}^{\cancel{2}}(2\omega_0 T_s)} = \frac{1}{\operatorname{tg}(2\omega_0 T_s)} = \cot(2\omega_0 T_s) = \operatorname{tg}\left(\frac{\pi}{2} - 2\omega_0 T_s\right)$$

$$\Rightarrow \varphi = \frac{\pi}{2} - 2w_0T_s + 2\pi \quad , \text{ uso } k=0$$

$$\boxed{\Rightarrow y_a[n] = 2\,\text{sen}(2w_0T_s)\cos\left(w_0 n T_s + \frac{\pi}{2} - 2w_0T_s\right)} = 2\,\text{sen}(2w_0T_s)\,\text{sen}(2w_0T_s - w_0 n T_s)$$

b) $x_b[n] = (1/2)^n\, u[n] = \begin{cases} 1/2^n & \text{si } n \geq 0 \\ 0 & \text{si } n < 0 \end{cases}$

$$X_b(w) = \sum_{n=-\infty}^{\infty} x_b[n] e^{-jwn} = \sum_{n=0}^{\infty} (1/2)^n e^{-jwn} = \sum_{n=0}^{\infty} \left(1/2\, e^{-jw}\right)^n \rightarrow \text{serie geométrica, cv si } \left|\tfrac{1}{2} e^{-jw}\right| < 1$$

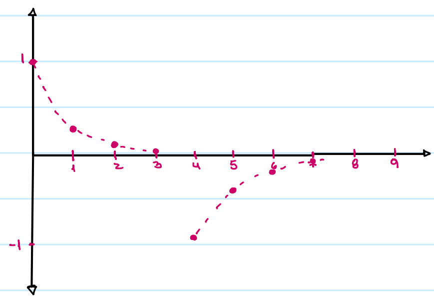
$\frac{1}{2} < 1 \ \checkmark \rightarrow$ cv

por serie geométrica: $X_b(w) = \dfrac{1}{1 - \frac{1}{2}e^{-jw}}$

$$Y_b(w) = \frac{1}{1 - 1/2\, e^{-jw}} \cdot \left(1 - e^{-4jw}\right) = \frac{1}{1 - 1/2\, e^{-jw}} - \frac{\overset{\text{corrimiento}}{e^{-4jw}}}{1 - 1/2\, e^{-jw}}$$

antitransformando:

$$\boxed{y_b[n] = \underbrace{(1/2)^n u[n]}_{\substack{0 \text{ para } n<0 \\ (1/2)^n \text{ para } n \geq 0}} - \underbrace{(1/2)^{n-4} u[n-4]}_{\substack{0 \text{ para } n<4 \\ -(1/2)^{n-4} \text{ para } n \geq 4}}} = \begin{cases} 0 & \text{para } n<0 \\ (1/2)^n & \text{para } 0 \leq n < 4 \\ (1/2)^n - (1/2)^{n-4} & \text{para } n \geq 4 \end{cases} \Rightarrow$$



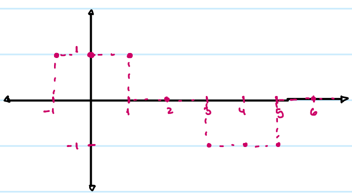
c) $x_c[n] = \underbrace{u[n+1]}_{\substack{0 \text{ para } n<-1 \\ 1 \text{ para } n \geq -1}} - \underbrace{u[n-2]}_{\substack{0 \text{ para } n<2 \\ 1 \text{ para } n \geq 2}} = \begin{cases} 0 & \text{si } n < -1 \\ 1 & \text{si } -1 \leq n < 2 \\ 0 & \text{si } n \geq 2 \end{cases}$

$$X_c(w) = \sum_{n=-\infty}^{\infty} x_c[n] e^{-jwn} = \sum_{n=-1}^{1} e^{-jwn} = e^{jw} + 1 + e^{-jw}$$

$$\Rightarrow Y_c(w) = \left(e^{jw} + 1 + e^{-jw}\right)\left(1 - e^{-4jw}\right)$$
$$= e^{jw} + 1 + e^{-jw} - \underbrace{e^{-4jw}}_{\text{corrimiento}}\left(e^{jw} + 1 + e^{-jw}\right)$$

antitransformando:

$$\boxed{y_c[n] = \underbrace{u[n+1]}_{\substack{0 \text{ para } n<-1 \\ 1 \text{ para } n \geq -1}} - \underbrace{u[n-2]}_{\substack{0 \text{ para } n<2 \\ 1 \text{ para } n \geq 2}} - \underbrace{u[n-3]}_{\substack{0 \text{ para } n<3 \\ 1 \text{ para } n \geq 3}} + \underbrace{u[n-6]}_{\substack{0 \text{ para } n<6 \\ 1 \text{ para } n \geq 6}}} = \begin{cases} 0 & \text{para } n < -1 \\ 1 & \text{para } -1 \leq n < 2 \\ 0 & \text{para } 2 \leq n < 3 \\ -1 & \text{para } 3 \leq n < 6 \\ 0 & \text{para } n \geq 6 \end{cases}$$



$\Rightarrow$ calcular las DFT de $y(n)$ para $N=8$

c) si quiero un corrimiento para $y'[n] = \overset{\text{como todo una posición a la derecha}}{y[n-1]}$

$$y_c[n] = u[n+1] - u[n-2] - u[n-3] + u[n-6] = \begin{cases} 0 & \text{para } n < -1 \\ 1 & \text{para } -1 \leq n < 2 \\ 0 & \text{para } 2 \leq n < 3 \\ -1 & \text{para } 3 \leq n < 6 \\ 0 & \text{para } n \geq 6 \end{cases}$$

$$\Rightarrow y_c'[n] = y_c[n-1] = u[n] - u[n-3] - u[n-4] + u[n-7] = \begin{cases} 0 & \text{para } n < 0 \\ 1 & \text{para } 0 \leq n < 3 \\ 0 & \text{para } 3 \leq n < 4 \\ -1 & \text{para } 4 \leq n < 7 \\ 0 & \text{para } n \geq 7 \end{cases}$$

$$Y_c'[k] = \sum_{n=0}^{7} y_c[n-1]\, e^{-j\frac{2\pi k n}{8}}$$

$n=0 \quad n=1 \quad n=2 \quad n=3 \quad n=4 \quad n=5 \quad n=6 \quad n=7$

$$Y'_c[\ell] = \sum_{n=0} y_c[n-1]\, e^{-j\frac{\quad}{8}}$$

$$= \underset{n=0}{1} + \underset{n=1}{e^{-j\frac{\ell\pi}{4}}} + \underset{n=2}{e^{-j\frac{\ell\pi}{2}}} + \underset{n=3}{0} - \underset{n=4}{e^{-j\ell\pi}} - \underset{n=5}{e^{-j\ell\pi\frac{5}{4}}} - \underset{n=6}{e^{-j\ell\pi\frac{3}{2}}} + \underset{n=7}{0}$$

para volver a $Y_c[\ell]$ aplico propiedad de corrimiento $\Rightarrow y_c[n] = y'_c[n+1]$

$$\Rightarrow Y_c[\ell] = e^{-j\frac{\pi\ell}{4}}\left[1 + e^{-j\frac{\ell\pi}{4}} + e^{-j\frac{\ell\pi}{2}} + 0 - e^{-j\ell\pi} - e^{-j\ell\pi\frac{5}{4}} - e^{-j\ell\pi\frac{3}{2}} + 0\right]$$

$$= \left[\cos\left(-\tfrac{\pi}{4}\ell\right) + j\,\mathrm{sen}\left(-\tfrac{\pi}{4}\ell\right)\right]\left(1 + \cos\left(-\tfrac{\pi}{4}\ell\right) + j\,\mathrm{sen}\left(-\tfrac{\pi}{4}\ell\right) + j\,\mathrm{sen}\left(-\tfrac{\ell\pi}{2}\right) - \cos(-\ell\pi) - \cos\left(-\tfrac{\ell\pi 5}{4}\right) - j\,\mathrm{sen}\left(-\tfrac{\ell\pi 5}{4}\right) - j\,\mathrm{sen}\left(\tfrac{\ell\pi 3}{2}\right)\right)$$

$$= \left(1 + 2\cos\left(\tfrac{\pi\ell}{4}\right) - \cos\left(\tfrac{3\pi\ell}{4}\right) - \cos(\pi\ell) - \cos\left(\tfrac{5\pi\ell}{4}\right)\right) + j\left(\mathrm{sen}\,\tfrac{3\pi\ell}{4} + \mathrm{sen}\,\tfrac{5\pi\ell}{4}\right)$$

$Y_c[0] = 0$

$Y_c[1] = 1 + \sqrt{2} + \frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} + j\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right)$

$= 2 + 2\sqrt{2}$

$Y_c[2] = \cancel{1} - \cancel{1} + j\left(-\cancel{1} + \cancel{1}\right) = 0$

$Y_c[3] = 1 - \sqrt{2} - \frac{\sqrt{2}}{2} + 1 - \frac{\sqrt{2}}{2} + j\left(\cancel{\frac{\sqrt{2}}{2}} - \cancel{\frac{\sqrt{2}}{2}}\right)$

$= 2 - 2\sqrt{2}$

$Y_c[4] = \cancel{1} - \cancel{2} + \cancel{1} - \cancel{1} + \cancel{1} = 0$

$Y_c[5] = 1 - \sqrt{2} - \frac{\sqrt{2}}{2} + 1 - \frac{\sqrt{2}}{2} + j\left(-\cancel{\frac{\sqrt{2}}{2}} + \cancel{\frac{\sqrt{2}}{2}}\right)$

$= 2 - 2\sqrt{2}$

$Y_c[6] = \cancel{1} - \cancel{1} + j\left(\cancel{1} - \cancel{1}\right) = 0$

$Y_c[7] = 1 + \sqrt{2} + \frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} + j\left(-\cancel{\frac{\sqrt{2}}{2}} + \cancel{\frac{\sqrt{2}}{2}}\right)$

$= 2 + 2\sqrt{2}$